

2 Appell's symbol and hypergeometric series

2.1 Introduction

The factorial function, $n! = 1 \cdot 2 \cdot 3 \cdots n$, is defined in the first instance when n is a positive integer, and the problem of giving $x!$ a useful meaning when x is any positive real number was solved in 1729 by Euler, who defined what is now called the gamma function. Chapters 2–4 are concerned in a sense with Euler's gamma function, but we begin with a simpler generalization of $n!$ called a shifted factorial and denoted here by Appell's symbol,

$$(a, n) = a(a+1)(a+2) \cdots (a+n-1). \quad (1)$$

This product of n factors, beginning with any complex number a and increasing by steps of unity, includes $(1, n) = n!$ as a special case. The product was studied by James Stirling (1730), but the present notation originated with Appell (1880a). Pochhammer's symbol $(a)_n$ is used more widely for the same quantity but is typographically inconvenient when n bears a subscript, as it often will in later chapters. Appell's symbol can be expressed in terms of the gamma function, which has more fundamental importance, but proceeding in the opposite direction requires a limiting process.

The value of Appell's symbol lies in its marvelous convenience in the study of hypergeometric series, which began in a systematic way with a memoir by Gauss (1812) on the series

$$1 + \frac{ab}{c}x + \frac{a(a+1)b(b+1)}{c(c+1)} \frac{x^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{(a, n)(b, n)}{(c, n)} \frac{x^n}{n!}, \quad (2)$$

where $|x| < 1$. The series is called hypergeometric because the geometric series

$$1 + x + x^2 + x^3 + \cdots \quad (3)$$

is the special case $a = 1, b = c$. Many important special functions are represented by the series (2). An elementary example is

$$\begin{aligned} \arcsin x &= x + \frac{x^3}{2 \cdot 3} + \frac{3x^5}{2 \cdot 4 \cdot 5} + \frac{3 \cdot 5x^7}{2 \cdot 4 \cdot 6 \cdot 7} + \cdots \\ &= x \sum_{n=0}^{\infty} \frac{(\frac{1}{2}, n)(\frac{1}{2}, n)}{(\frac{3}{2}, n)} \frac{(x^2)^n}{n!}, \quad |x| < 1. \end{aligned} \quad (4)$$

In general a power series $\sum_{n=0}^{\infty} A_n x^n$ is called a hypergeometric series if the ratio A_{n+1}/A_n is a rational function of n . The sum of the series inside its circle of convergence is called a hypergeometric function, and the same name is used for its analytic continuation outside the circle of convergence. Hypergeometric functions and their generalizations have done more than anything else to bring order and unity into the subject of special functions.

2.2 Appell's symbol

We assumed tacitly in (2.1-1) that the integer n is positive and in (2.1-2) that $(a, 0) = 1$. This value of $(a, 0)$ is a natural choice, for (2.1-1) implies $(a, n) = (a, n-1)(a+n-1)$ and hence,

$$(a, n-1) = \frac{(a, n)}{a+n-1}, \quad a+n-1 \neq 0, \quad (1)$$

for $n = 2, 3, 4, \dots$. Whenever (a, n) has already been defined but $(a, n-1)$ has not, it is sensible to choose a definition that preserves (1). Accordingly, in the case $n = 1$, we shall define $(a, 0)$ to be $(a, 1)/a = 1$ provided $a \neq 0$. To make $(a, 0)$ continuous in a , we define $(0, 0) = 1$ also. Setting $n = 0$ in (1), we then define $(a, -1) = 1/(a-1)$, $a \neq 1$, and so forth.

DEFINITION 2.2-1 Let $n-1 \in \mathbb{N}$ and $a \in \mathbb{C}$. (The sets $\mathbb{N}$ and $\mathbb{C}$ are defined in Appendix A.) Define

$$\begin{aligned} (a, n) &= a(a+1)(a+2) \cdots (a+n-1), \\ (a, 0) &= 1, \\ (a, -n) &= \frac{1}{(a-1)(a-2) \cdots (a-n)}, \quad a \neq 1, 2, \dots, n. \end{aligned} \quad (2)$$

Note that $(-n, m) = 0$ if m and n are integers and $0 \leq n < m$. For example $(-2, 4) = (-2)(-1)(0)(1) = 0$. As a result a series sometimes terminates after a finite number of terms. For instance the binomial series is

$$(1+x)^v = 1 + vx + v(v-1)\frac{x^2}{2!} + \cdots, \quad |x| < 1, \quad (3)$$

or alternatively, in a form to be used often,

$$(1-x)^{-a} = 1 + ax + a(a+1)\frac{x^2}{2!} + \cdots = \sum_{m=0}^{\infty} (a, m) \frac{x^m}{m!}, \quad |x| < 1. \quad (4)$$

If $a = -n$ is a negative integer, all coefficients $(-n, m)$ with $m > n$ vanish, the binomial series terminates, and (4) reduces to the binomial theorem.

Many identities involving Appell symbols can be proved directly from the definition by writing each symbol as a product of factors or, more economically, by using the three identities that follow. Other identities are given in the Exercises at the end of the chapter, but only these three are worth remembering. The second and third foreshadow important properties of the gamma function.

THEOREM 2.2-2 Let the complex number a and the integers m and n be such that both sides of the following equations are well defined. Then

$$(a, m+n) = (a, m)(a+m, n) \quad (\text{addition formula}), \quad (5)$$

$$(a, -n) = (-1)^n / (1-a, n) \quad (\text{reflection formula}), \quad (6)$$

$$(2a, 2n) = 2^{2n} (a, n) (a + \frac{1}{2}, n) \quad (\text{duplication formula}). \quad (7)$$

Proof All cases with $n = 0$ or $m = 0$ are trivial. The proof of (6) for $n > 0$ is

$$\begin{aligned} (a, -n) &= \frac{1}{(a-1)(a-2)\cdots(a-n)} = \frac{(-1)^n}{(1-a)(2-a)\cdots(n-a)} \\ &= \frac{(-1)^n}{(1-a, n)}. \end{aligned}$$

If $n < 0$, we write $n = -N$ and apply the preceding result to the right-hand side of (6):

$$\frac{(-1)^n}{(1-a, n)} = \frac{(-1)^{-N}}{(1-a, -N)} = (1-1+a, N) = (a, -n).$$

If $n > 0$, (7) is proved by grouping together alternate factors of

$$\begin{aligned} (2a, 2n) &= 2a(2a+1)(2a+2)(2a+3) \cdots (2a+2n-2)(2a+2n-1) \\ &= [2a(2a+2) \cdots (2a+2n-2)][(2a+1)(2a+3) \cdots (2a+2n-1)] \\ &= 2^{2n}[a(a+1) \cdots (a+n-1)][(a+\frac{1}{2})(a+\frac{3}{2}) \cdots (a+n-\frac{1}{2})] \\ &= 2^{2n}(a, n)(a+\frac{1}{2}, n). \end{aligned}$$

If $n = -N < 0$, we use the preceding result together with (6):

$$\begin{aligned} (2a, 2n) &= (2a, -2N) = \frac{1}{(1-2a, 2N)} = \frac{1}{2^{2N}(\frac{1}{2}-a, N)(1-a, N)} \\ &= \frac{2^{2n}}{(\frac{1}{2}-a, -n)(1-a, -n)} = 2^{2n}(\frac{1}{2}+a, n)(a, n). \end{aligned}$$

We prove (5) first for the cases with $m > 0$. If also $n > 0$, then

$$\begin{aligned} (a, m+n) &= [a(a+1) \cdots (a+m-1)] \\ &\quad \times [(a+m)(a+m+1) \cdots (a+m+n-1)] \\ &= (a, m)(a+m, n). \end{aligned}$$

If $n = -N < 0$ and $m > N$, then

$$\begin{aligned} (a, m+n) &= (a, m-N) \\ &= [a(a+1) \cdots (a+m-N-1)] \frac{(a+m-N) \cdots (a+m-1)}{(a+m-N) \cdots (a+m-1)} \\ &= \frac{(a, m)}{(a+m-1) \cdots (a+m-N)} \\ &= (a, m)(a+m, -N) = (a, m)(a+m, n). \end{aligned}$$

If $m = N$, this proof remains unchanged, except that the quantity in square brackets is replaced by $(a, 0) = 1$. If $m < N$,

$$\begin{aligned} (a, m+n) &= (a, m-N) \\ &= \frac{1}{(a-1)(a-2) \cdots (a+m-N)} \frac{a(a+1) \cdots (a+m-1)}{a(a+1) \cdots (a+m-1)} \\ &= \frac{(a, m)}{(a+m-1) \cdots a(a-1) \cdots (a+m-N)} \\ &= (a, m)(a+m, -N) = (a, m)(a+m, n). \end{aligned}$$

Finally, if $m < 0$, we write $m = -M$ and use the preceding results together with (6):

$$\begin{aligned} (a, m+n) &= (a, -M+n) = \frac{(-1)^{-M+n}}{(1-a, M-n)} = \frac{(-1)^{-M+n}}{(1-a, M)(1-a+M, -n)} \\ &= (a, -M)(a-M, n) = (a, m)(a+m, n). \quad \square \end{aligned}$$

An important example of the use of (5) and (6) occurs in connection with sums of the type $\sum_{m=0}^n f(m, n)$. A factor in f having the form $1/(a, m)$ or $1/(a, n-m)$ can be changed into $(-1)^m(b, n-m)$ or $(-1)^m(b, m)$, respectively, multiplied by a quantity independent of m . The more obvious case is

$$\frac{1}{(a, n-m)} = \frac{1}{(a, n)(a+n, -m)} = \frac{(-1)^m(1-a-n, m)}{(a, n)}, \quad (8)$$

where $b = 1-a-n$ does not involve the summation index. In the other case it is a useful trick to add and subtract n :

$$\begin{aligned} \frac{1}{(a, m)} &= \frac{1}{(a, n-n+m)} = \frac{1}{(a, n)(a+n, -n+m)} \\ &= \frac{(-1)^{n-m}(1-a-n, n-m)}{(a, n)}. \end{aligned} \quad (9)$$

Although (9) is the same as (8) with m replaced by $n-m$, it is usually quicker to follow the procedure we have just used than to look up either result.

In discussing questions of convergence, it will be useful to have an inequality for Appell's symbol. If $n-1 \in \mathbb{N}$, then

$$\begin{aligned} |(a, n)| &= |a| \cdot |a+1| \cdots |a+n-1| \\ &\leq |a| \cdot (|a|+1) \cdots (|a|+n-1) = (|a|, n). \end{aligned}$$

Since the case $n=0$ is trivial, we have

$$|(a, n)| \leq (|a|, n), \quad n \in \mathbb{N}. \quad (10)$$

Appell's symbol is related to the binomial coefficient, defined for $n \in \mathbb{N}$ and $a \in \mathbb{C}$ by

$$\binom{a}{0} = 1, \quad \binom{a}{n} = \frac{a(a-1) \cdots (a-n+1)}{n!}, \quad n > 0. \quad (11)$$

It follows immediately from (2) that

$$\binom{a}{n} = (-1)^n \frac{(-a, n)}{n!}. \quad (12)$$

If n is a negative integer, $1/n!$ is defined to be 0 in order to preserve the relation $1/(m-1)! = m/m!$ for every $m \in \mathbb{Z}$. Therefore, although (a, n) remains useful if n is a negative integer, $\binom{a}{n}$ can only be taken to vanish and is not usually defined.

2.3 Vandermonde's theorem

An important summation formula was proved by Vandermonde (1772). Although the Chinese mathematician Chu published a less general form of the theorem (Exercise 2.3-12) in 1303 [see Askey (1975b, pp. 59–60)], we will call it Vandermonde's theorem and for convenience will use the same name to designate an extension to multiple sums. Many authors refer to a variant of the formula (Exercise 2.3-1) as Vandermonde's theorem or lemma.

THEOREM 2.3-1 Let $n \in \mathbb{N}$ and $a, b \in \mathbb{C}$. Then

$$\sum_{m=0}^n \frac{(a, m)(b, n-m)}{m!(n-m)!} = \frac{(a+b, n)}{n!}. \quad (1)$$

Proof Substitute the binomial series (2.2-4) in

$$(1-t)^{-a}(1-t)^{-b} = (1-t)^{-a-b}. \quad (2)$$

This yields

$$\sum_{m=0}^{\infty} (a, m) \frac{t^m}{m!} \sum_{m'=0}^{\infty} (b, m') \frac{t^{m'}}{m'!} = \sum_{n=0}^{\infty} (a+b, n) \frac{t^n}{n!}. \quad (3)$$

Equate the coefficients of t^n :

$$\sum \frac{(a, m)(b, m')}{m!m'!} = \frac{(a+b, n)}{n!}, \quad (4)$$

where the summation extends over all nonnegative integers m and m' whose sum is n . Since $m' = n - m$ in all terms, the sum is the same as that in (1). $\square$

Instead of taking a product of two binomial series, we could have differentiated (2) n times and set $t = 0$ to prove (4). This alternative method produces a general rule: If one wants only to equate coefficients of a certain power in a relation between power series, questions of convergence are irrelevant [see Gould (1974)]. Under such circumstances we shall call an equation like (3) an equality between formal power series which may or may not converge.

An extension of (1) to multiple sums is easy.

2.3 Vandermonde's theorem

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THEOREM 2.3-2 If $b_1, b_2, \dots, b_k$ are any complex numbers, then

$$\sum \frac{(b_1, m_1) \cdots (b_k, m_k)}{m_1! \cdots m_k!} = \frac{(b_1 + \cdots + b_k, n)}{n!}, \quad (5)$$

where the summation extends over all nonnegative integers $m_1, m_2, \dots, m_k$ whose sum is n .

Proof Substituting the binomial series as a formal power series in

$$(1 - t)^{-b_1} \cdots (1 - t)^{-b_k} = (1 - t)^{-(b_1 + \cdots + b_k)}$$

produces

$$\sum_{m_1=0}^{\infty} (b_1, m_1) \frac{t^{m_1}}{m_1!} \cdots \sum_{m_k=0}^{\infty} (b_k, m_k) \frac{t^{m_k}}{m_k!} = \sum_{n=0}^{\infty} (b_1 + \cdots + b_k, n) \frac{t^n}{n!}.$$

Equate the coefficients of t^n to prove (5). $\square$

Vandermonde's theorem contains as limiting cases the binomial and multinomial theorems. In (1) we set $a = cx$ and $b = cy$, divide both sides by c^n , and let $c \rightarrow \infty$. Since

$$\frac{(cx, m)}{c^m} = \frac{cx(cx+1) \cdots (cx+m-1)}{c^m} = x \left(x + \frac{1}{c} \right) \cdots \left(x + \frac{m-1}{c} \right) \rightarrow x^m \quad (6)$$

and similarly, $(cy, n-m)/c^{n-m} \rightarrow y^{n-m}$, we obtain

$$\sum_{m=0}^n \frac{x^m y^{n-m}}{m! (n-m)!} = \frac{(x+y)^n}{n!}. \quad (7)$$

This is equivalent to the usual form of the binomial theorem,

$$(x+y)^n = \sum_{m=0}^n \binom{n}{m} x^m y^{n-m}. \quad (8)$$

Likewise, setting $b_i = cx_i$, $i = 1, \dots, k$, in (5), dividing both sides by c^n , and letting $c \rightarrow \infty$, we obtain the multinomial theorem,

$$(x_1 + \cdots + x_k)^n = \sum \frac{n!}{m_1! \cdots m_k!} x_1^{m_1} \cdots x_k^{m_k}, \quad (9)$$

where the summation extends over all nonnegative integers $m_1, \dots, m_k$ whose sum is n .

Vandermonde's theorem is related also to probability distributions. Dividing (1) by its right-hand side, we have

$$\sum_{m=0}^n p(m) = 1, \quad p(m) = \binom{n}{m} \frac{(a, m)(b, n-m)}{(a+b, n)}, \quad (a+b, n) \neq 0. \quad (10)$$

We can interpret $p(m)$ as a probability of finding the integer m provided $p(m)$ is real and nonnegative for $m = 0, 1, \dots, n$. This is obviously true in the case $a > 0, b > 0$, and the limit discussed above leads to the binomial distribution in this case. It is true also if $a = -N_1$ and $b = -N_2$, where N_1 and N_2 are positive integers with $N_1 + N_2 \geq n$. This case is known as the hypergeometric distribution because its generating function is a hypergeometric function of the form (2.1-2). Suppose n balls are chosen at random from a population consisting of N_1 red balls and N_2 black balls. The probability that the sample contains exactly m red balls is $p(m)$. Populations containing more than two colors lead to a distribution which is similarly connected with (5). These distributions are important in the statistical theory of industrial quality control, card games, and estimation of animal populations [see Feller (1968) and Wilks (1962)].

2.4 Hypergeometric series

A hypergeometric series is a series $\sum u_m$ such that u_{m+1}/u_m is a rational function of m . The series of Gauss (1812) is

$$1 + \frac{ab}{c}z + \frac{a(a+1)b(b+1)}{c(c+1)}\frac{z^2}{2!} + \dots = \sum_{m=0}^{\infty} \frac{(a, m)(b, m)}{(c, m)} \frac{z^m}{m!}, \quad |z| < 1. \quad (1)$$

To indicate that it has two parameters (a and b) in the numerator of each term and one parameter (c) in the denominator, the series (1) is called a ${}_2F_1$ series:

$${}_2F_1(a, b; c; z) = \sum_{m=0}^{\infty} \frac{(a, m)(b, m)}{(c, m)} \frac{z^m}{m!}, \quad |z| < 1. \quad (2)$$

We assume $c \neq 0, -1, -2, \dots$ to prevent the denominators from vanishing. If the series does not terminate and if its general term is denoted by u_m , then

$$\frac{u_{m+1}}{u_m} = \frac{(a+m)(b+m)z}{(c+m)(1+m)} = \frac{(1+a/m)(1+b/m)z}{(1+c/m)(1+1/m)}, \quad (3)$$

and hence

$$\lim_{m \rightarrow \infty} \left| \frac{u_{m+1}}{u_m} \right| = |z|. \quad (4)$$

By the ratio test the series converges absolutely if $|z| < 1$ and diverges if $|z| > 1$. At a given point z on the unit circle the series may converge or diverge depending on the values of z and the parameters.

In the ${}_2F_1$ notation (2.1-4) becomes

$$\arcsin z = z {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; \frac{3}{2}; z^2\right), \quad |z| < 1. \quad (5)$$

Some other examples are

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = {}_2F_1(1, 1; 2; -z), \quad |z| < 1, \quad (6)$$

$$\begin{aligned} \operatorname{arctanh} z &= \frac{1}{2} \log \frac{1+z}{1-z} = z + \frac{z^3}{3} + \frac{z^5}{5} + \cdots \\ &= z {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; z^2\right), \quad |z| < 1, \end{aligned} \quad (7)$$

$$P_n(z) = {}_2F_1\left(-n, 1+n; 1; \frac{1-z}{2}\right), \quad n \in \mathbb{N}, \quad (8)$$

where P_n is the Legendre polynomial of degree n (see Section 6.7) and $\log$ means the natural logarithm. In (8) the ${}_2F_1$ series terminates because one of the numerator parameters is a nonpositive integer and the corresponding Appell symbol $(-n, m)$ is zero for $m > n$. A terminating ${}_2F_1$ series is called a ${}_2F_1$ polynomial.

Clausen (1828) introduced a series similar to (1) with three numerator parameters and two denominator parameters. It is now called a ${}_3F_2$ series, and a series with p numerator parameters $(a) = (a_1, \dots, a_p)$ and q denominator parameters $(c) = (c_1, \dots, c_q)$ is called a ${}_pF_q$ series. Several notations in common use are

$$\begin{aligned} {}_pF_q[(a), (c), z] &= {}_pF_q(a_1, \dots, a_p; c_1, \dots, c_q; z) \\ &= {}_pF_q\left[\begin{matrix} a_1, \dots, a_p \\ c_1, \dots, c_q \end{matrix}; z\right] = \sum_{m=0}^{\infty} \frac{(a_1, m) \cdots (a_p, m)}{(c_1, m) \cdots (c_q, m)} \frac{z^m}{m!}. \end{aligned} \quad (9)$$

We assume $c_1, \dots, c_q \in U$, where the set U is defined in Appendix A. Denoting the general term by u_m , we have

$$\frac{u_{m+1}}{u_m} = \frac{(a_1 + m) \cdots (a_p + m)z}{(c_1 + m) \cdots (c_q + m)(1 + m)} = \frac{m^{p-q-1}(1 + a_1/m) \cdots (1 + a_p/m)z}{(1 + c_1/m) \cdots (1 + c_q/m)(1 + 1/m)}. \quad (10)$$

Provided that the series does not terminate, the ratio test shows if $p = q + 1$, the series converges absolutely inside the unit circle and diverges outside, as we found earlier for the ${}_2F_1$ series. Another example with $p = q + 1$ is the binomial series,

$$(1-z)^{-a} = \sum_{m=0}^{\infty} (a, m) \frac{z^m}{m!} = {}_1F_0(a; z), \quad |z| < 1. \quad (11)$$

If $p < q + 1$, then $|u_{m+1}/u_m| \rightarrow 0$ as $m \rightarrow \infty$. The series converges absolutely at every point of the finite z plane (denoted by $\mathbb{C}$) and therefore

represents an entire function, i.e. a function that is holomorphic on $\mathbb{C}$. Three elementary examples are

$$e^z = \sum_{m=0}^{\infty} \frac{z^m}{m!} = {}_0F_0(z), \quad (12)$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots = {}_0F_1\left(\frac{1}{2}; -\frac{z^2}{4}\right), \quad (13)$$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots = {}_0F_1\left(\frac{3}{2}; -\frac{z^2}{4}\right), \quad (14)$$

all valid for $z \in \mathbb{C}$. Two other important examples are the Bessel function

$$\begin{aligned} J_0(z) &= 1 - \frac{z^2}{2^2} + \frac{z^4}{2^2 \cdot 4^2} - \frac{z^6}{2^2 \cdot 4^2 \cdot 6^2} + \cdots \\ &= \sum_{m=0}^{\infty} \frac{(-z^2/4)^m}{(m!)^2} = {}_0F_1\left(1; -\frac{z^2}{4}\right), \quad z \in \mathbb{C}, \end{aligned} \quad (15)$$

and the error function

$$\begin{aligned} \operatorname{Erf}(z) &= \int_0^z e^{-t^2} dt = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \frac{z^{2m+1}}{2m+1} \\ &= {}_1F_1\left(\frac{1}{2}; \frac{3}{2}; -z^2\right), \quad z \in \mathbb{C}. \end{aligned} \quad (16)$$

The error function is a special case of the confluent hypergeometric series studied by Kummer (1836),

$${}_1F_1(a; c; z) = 1 + \frac{a}{c} z + \frac{a(a+1)}{c(c+1)} \frac{z^2}{2!} + \cdots, \quad c \in U, \quad z \in \mathbb{C}. \quad (17)$$

A ${}_1F_1$ polynomial is called a Laguerre polynomial, often denoted by

$$L_n^\alpha(z) = \frac{(1+\alpha, n)}{n!} {}_1F_1(-n; 1+\alpha; z), \quad 1+\alpha \in U. \quad (18)$$

If $p > q + 1$, the ratio test shows that the ${}_pF_q$ series diverges for all $z \neq 0$ unless it terminates. The most important example of a terminating series of this kind is the Hermite polynomial

$$H_n(z) = (2z)^n {}_2F_0\left(\frac{-n}{2}, \frac{1-n}{2}; -\frac{1}{z^2}\right), \quad n \in \mathbb{N}. \quad (19)$$

Regardless of whether n is even or odd, one of the two numerator parameters is a nonpositive integer.

Since ${}_pF_q$ series have been studied extensively, it is often advantageous to express a given series, if possible, in the standard form (9). This can

be done only if the ratio u_{m+1}/u_m of successive terms is a rational function of m , as it is in (10). If the series $S = \sum_{m=0}^{\infty} u_m$ does satisfy

$$\frac{u_{m+1}}{u_m} = \frac{(a_1 + m) \cdots (a_p + m)z}{(c_1 + m) \cdots (c_q + m)(1 + m)} \quad (20)$$

for all m , where none of the numbers $a_1, \dots, c_q$ is zero or a negative integer, then

$$S = u_0 {}_pF_q[(a), (c), z]. \quad (21)$$

Note that the factor $(1 + m)$ in the denominator of (20) can be supplied if necessary by multiplication with $(1 + m)/(1 + m)$; one of the numerator parameters will then be unity. If the series S terminates at the term $u_n \neq 0$ and if (20) holds for $m = 0, 1, \dots, n$, then one of the a_i is $-n$ and (21) is still true. If (20) holds for $m = n + 1, n + 2, \dots$ and one of the c_i is $-n$, we can define $v_m = u_{n+m}$ and express $\sum_{m=0}^{\infty} v_m$ in terms of a ${}_pF_q$ series.

EXAMPLE 2.4-1 Express as a ${}_pF_q$ series the series

$$S = z + \frac{1}{3 \cdot 5} z^3 + \frac{1 \cdot 3}{5 \cdot 7 \cdot 9} z^5 + \frac{1 \cdot 3 \cdot 5}{7 \cdot 9 \cdot 11 \cdot 13} z^7 + \cdots \quad (22)$$

We have

$$u_m = \frac{1 \cdot 3 \cdots (2m-1)z^{2m+1}}{(2m+1)(2m+3) \cdots (4m+1)}$$

and hence

$$\frac{u_{m+1}}{u_m} = \frac{(2m+1)^2 z^2}{(4m+3)(4m+5)} = \frac{(\frac{1}{2} + m)^2}{(\frac{3}{4} + m)(\frac{5}{4} + m)} \frac{(1+m)z^2}{(1+m)4}, \quad m \in \mathbb{N}.$$

Since $u_0 = z$,

$$S = z {}_3F_2(\frac{1}{2}, \frac{1}{2}, 1; \frac{3}{4}, \frac{5}{4}; z^2/4). \quad (23)$$

The ${}_pF_q$ notation provides a standard form by which to classify a remarkable number of familiar special functions like the ones used earlier as examples. If an unfamiliar function is found to have a series expansion which can be put in this standard form, one may be able to identify it at once, for instance as a Bessel function. In more complicated cases one will at least know where to look for information about functions of the same type and will sometimes find that a substantial amount of general theory is available.

Some information about the simpler types of ${}_pF_q$ series is listed here for orientation and convenient reference.

${}_0F_0$ The exponential function.

${}_1F_0$ The binomial series.

${}_0F_1$ Every ${}_0F_1$ series is a Bessel function. It can always be expressed as a restricted ${}_1F_1$ series (i.e. a ${}_1F_1$ series with the values of the parameters

restricted so that not both of them can be chosen freely). Special cases include the circular and hyperbolic sine and cosine functions.

${}_1F_1$ and ${}_2F_0$ The confluent hypergeometric series or Kummer's series (${}_1F_1$) is inextricably connected with the ${}_2F_0$ series, which diverges unless it terminates but is an asymptotic expansion of a well-defined function (Section 5.12). These functions are equivalent to confluent hypergeometric functions defined by Whittaker and Tricomi. Their special cases include the error functions, incomplete gamma functions, exponential and logarithmic integrals, cylinder (Bessel, Neumann, Hankel, and Macdonald) functions, and parabolic cylinder functions. Cases in which the series terminate include the Hermite, Laguerre, and Bessel polynomials (Exercises 2.4-3 and 3.11-1).

${}_2F_1$ The ordinary or Gauss hypergeometric series. Its special cases include logarithms, inverse circular and hyperbolic functions, Legendre polynomials and functions, associated Legendre functions, Chebyshev polynomials, Gegenbauer or ultraspherical polynomials, Jacobi polynomials, incomplete beta functions, and complete elliptic integrals of the first and second kinds.

These six cases, in which $p = 0, 1$, or 2 and $q = 0$ or 1 , occur often in applied mathematics, primarily because the functions satisfy linear differential equations of order two or less. A ${}_pF_q$ series satisfies a linear equation (Exercise 2.4-6) of order $\max\{p, q + 1\}$. Fourth-order differential equations are common in the theory of elasticity, and there one meets ${}_0F_3$, ${}_2F_3$, and ${}_4F_3$, for example. Some cases of particular interest are

${}_1F_2$ Special cases include the Fresnel integrals (which occur in optics), the integral sine, Struve functions, and Lommel functions.

${}_3F_2$ Clausen's series. Several theorems give the sums of restricted ${}_3F_2$ series with unit argument. In quantum mechanics the vector addition coefficients can be written as terminating ${}_3F_2$ series with unit argument [see Rose (1955, Appendix B)].

${}_0F_3$ Special cases include Kelvin's ber and bei functions (see Exercise 2.4-8) which can also be expressed in terms of ${}_0F_1$ series. The Kelvin functions occur in the theory of the skin effect in electrical engineering.

${}_2F_3$ The product of two ${}_0F_1$ series (Bessel functions) is a restricted ${}_2F_3$ series (Exercise 2.4-1). The integral cosine is a special ${}_2F_3$ series.

${}_4F_3$ Some restricted series with argument 1 or -1 can be summed in terms of gamma functions. In quantum mechanics the Racah coefficients are terminating ${}_4F_3$ series with unit argument [see Rose (1955, Appendix B)].

Although the notion of a ${}_pF_q$ series serves to classify an impressive variety of special functions, one must not expect too much. The most

important special function, aside from the elementary ones, is the gamma function, and unlike the ${}_pF_q$ series it satisfies no simple differential equation. Some important classes of special functions satisfying linear differential equations with periodic coefficients [see Arscott (1964)] are not of ${}_pF_q$ type. The properties of these functions tend to be thorny, and except for the Mathieu functions, they have not been studied as thoroughly as hypergeometric series. Elliptic functions have beautiful properties but satisfy nonlinear differential equations. Although not ${}_pF_q$ series, they are closely related by way of a generalization known as a basic hypergeometric series. Moreover, they are the functions inverse to the elliptic integrals (as the exponential is inverse to the logarithm), and the elliptic integrals are hypergeometric series in two or more variables.

A double series,

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} A(m, n) x^m y^n, \quad (24)$$

is called hypergeometric if the ratios $A(m+1, n)/A(m, n)$ and $A(m, n+1)/A(m, n)$ are rational functions of m and n . A classification now requires more than two integers p and q because the numerator or denominator parameters no longer need to be all on the same footing. For example, in the double hypergeometric series

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(a, m+n)(b, m)(b', n)}{(c, m+n)} \frac{x^m y^n}{m! n!}, \quad (25)$$

the parameter a clearly has a different role from b or b' . This is only one of four double series known as Appell functions, but it includes all elliptic integrals of the first and second kinds as well as all complete elliptic integrals of the third kind (Carlson, 1961).

As a starting point for systematic theory, an infinite series is sometimes less satisfactory than an integral representation. Even the general theory of ${}_pF_q$ series is nowadays founded on representation by contour integrals with integrands formed from gamma functions. Elliptic integrals arise naturally from integration and only seldom from the double series (25). Moreover, there is at present no acceptable classification of double hypergeometric series because the one in use since 1889 has proved to be faulty (Carlson, 1976b).

After discussing the gamma function, we shall be concerned primarily with ${}_pF_q$ series having $p = 0, 1$, or 2 and $q = 0$ or 1 , with elliptic integrals, and with some multiple series like (25) in several variables. The functions will be defined by integral representations rather than series representations, and the ${}_pF_q$ notation will have a useful but secondary role.

2.5 Wallis' formula for π

The Appell symbol $(\frac{1}{2}, n)$ is a rational number for every integer n , but in the limit of large n it has two remarkable connections with the transcendental numbers π and e . The first of these connections was found by John Wallis at Oxford in 1656, before the invention of the integral calculus. However, we shall use calculus to prove Wallis' formula,

$$\frac{\pi}{2} = \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdots}, \quad (1)$$

which can be written in terms of Appell symbols as

$$\frac{\pi}{2} = \lim_{n \rightarrow \infty} \frac{(n!)^2}{(\frac{1}{2}, n)(\frac{3}{2}, n)}. \quad (2)$$

We shall connect $(\frac{1}{2}, n)$ with e in Section 2.7.

THEOREM 2.5-1 Let $\pi/2$ be the first positive root of the equation $\cos \theta = 0$. Then

$$\lim_{n \rightarrow \infty} \frac{(\frac{1}{2}, n)n^{1/2}}{(1, n)} = \pi^{-1/2}. \quad (3)$$

Proof We define

$$A_k = \int_0^{\pi/2} \sin^k \theta \, d\theta, \quad k \in \mathbb{N}. \quad (4)$$

Then $A_0 = \pi/2$ and $A_1 = 1$. For $k \geq 2$ a recurrence formula can be obtained by integration by parts:

$$\begin{aligned} A_k &= [-\sin^{k-1} \theta \cos \theta]_0^{\pi/2} + (k-1) \int_0^{\pi/2} \sin^{k-2} \theta \cos^2 \theta \, d\theta \\ &= (k-1)(A_{k-2} - A_k), \end{aligned}$$

and thus

$$A_k = \frac{k-1}{k} A_{k-2}, \quad k = 2, 3, 4, \dots$$

Hence, if n is a positive integer,

$$\begin{aligned} A_{2n} &= \frac{2n-1}{2n} \frac{2n-3}{2n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} A_0 = \frac{(\frac{1}{2}, n) \pi}{n!}, \\ A_{2n+1} &= \frac{2n}{2n+1} \frac{2n-2}{2n-1} \cdots \frac{4}{5} \cdot \frac{2}{3} A_1 = \frac{n!}{(\frac{3}{2}, n)}. \end{aligned} \quad (5)$$

Because $0 < \sin \theta < 1$ if $0 < \theta < \pi/2$, it is plain from the definition that A_k is a strictly decreasing function of k :

$$A_{2n+1} < A_{2n} < A_{2n-1}.$$

It follows that

$$\frac{n!}{(\frac{3}{2}, n)} < \frac{(\frac{1}{2}, n) \pi}{n!} < \frac{(n-1)!}{(\frac{3}{2}, n-1)} \quad (6)$$

and hence, for some number ξ (depending on n) between 0 and 1,

$$\frac{\pi}{2} = \frac{(n!)^2}{(\frac{1}{2}, n)(\frac{3}{2}, n)} \left(1 + \frac{\xi}{2n}\right), \quad 0 < \xi < 1. \quad (7)$$

This implies Wallis' formula (2). A slight rearrangement of (2) shows that

$$\pi = \lim_{n \rightarrow \infty} \frac{(n!)^2}{(\frac{1}{2}, n)^2 n} \cdot \frac{n}{n + \frac{1}{2}} = \lim_{n \rightarrow \infty} \frac{(n!)^2}{(\frac{1}{2}, n)^2 n},$$

which implies (3). $\square$

2.6 Arithmetic, geometric, and logarithmic means

If the real numbers x and y are positive and unequal, their arithmetic, geometric, and logarithmic means are $\frac{1}{2}(x+y)$, $(xy)^{1/2}$, and $(x-y)/(\log x - \log y)$, respectively. Since $x+y-2(xy)^{1/2} = (x^{1/2} - y^{1/2})^2 > 0$, the arithmetic mean is larger than the geometric mean. We shall prove that the logarithmic mean lies between them.

THEOREM 2.6-1 Let x and y be positive numbers. If $x \neq y$, then

$$(xy)^{1/2} < \frac{x-y}{\log x - \log y} < \frac{x+y}{2}. \quad (1)$$

Proof For every $t > 0$ the inequality of the arithmetic and geometric means implies

$$t^2 + (x+y)t + [\frac{1}{2}(x+y)]^2 > t^2 + (x+y)t + xy > t^2 + 2(xy)^{1/2}t + xy.$$

Hence

$$\int_0^\infty \frac{dt}{(t + \frac{1}{2}(x+y))^2} < \int_0^\infty \frac{dt}{(t+x)(t+y)} < \int_0^\infty \frac{dt}{(t+(xy)^{1/2})^2}.$$

Integrating the second member by the method of partial fractions, we find

$$\frac{2}{x+y} < \lim_{R \rightarrow \infty} \left[\frac{\log(t+y) - \log(t+x)}{x-y} \right]_0^R < (xy)^{-1/2},$$

which is equivalent to (1). $\square$

The inequalities in (1) become sharper as the relative difference of x and y tends to zero. The ratio of the arithmetic and logarithmic means tends to unity at a rate which is constrained by the following inequalities.

THEOREM 2.6-2 Let $x, y \in \mathbb{R}_>$ and assume $x \neq y$. Then

$$1 + \frac{(x-y)^2}{3(x+y)^2} < \frac{x+y}{2} \frac{\log x - \log y}{x-y} < 1 + \frac{(x-y)^2}{12xy}. \quad (2)$$

Proof We define

$$A = \frac{x+y}{2} \frac{\log x - \log y}{x-y} - 1 \quad \text{and} \quad z = \frac{x-y}{x+y}. \quad (3)$$

Then $0 < |z| < 1$ and

$$A = \frac{1}{2z} \log \frac{1+z}{1-z} - 1 = \frac{1}{z} \operatorname{arctanh} z - 1 = \frac{z^2}{3} + \frac{z^4}{5} + \frac{z^6}{7} + \cdots. \quad (4)$$

Hence

$$\frac{z^2}{3} < A < \frac{z^2}{3} (1 + z^2 + z^4 + \cdots) = \frac{z^2}{3(1-z^2)},$$

which is equivalent to (2). $\square$

Setting $y = x + 1$ in (2), we find, for $x > 0$,

$$\left(x + \frac{1}{2}\right) \log \frac{x+1}{x} = 1 + \frac{1}{12(x^2 + x + \theta)}, \quad (5)$$

$$\left(\frac{x+1}{x}\right)^{x+1/2} = \exp \left[1 + \frac{1}{12(x^2 + x + \theta)} \right], \quad (6)$$

where θ depends on x but satisfies $0 < \theta < \frac{1}{4}$. We shall use this result to prove Stirling's formula in Section 2.7. Incidentally, (6) shows that

$$e = \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x. \quad (7)$$

In the theory of the gamma function we shall meet the quantity A , defined by (3), in a context where x and y are complex (Section 3.8).

THEOREM 2.6-3 Let $x, y \in \mathbb{C}$ and assume $0 < |x-y| < |x+y|$. Then

$$\left| \frac{x+y}{2} \frac{\log x - \log y}{x-y} - 1 \right| < \frac{1}{3} \frac{|x-y|^2}{|x+y|^2 - |x-y|^2}, \quad (8)$$

where $\log x - \log y$ is taken to be the principal value of $\log(x/y)$.

Remark The condition $0 < |x - y| < |x + y|$ means that 0 lies outside the circle constructed on the line segment joining x and y as diameter, because the radius of this circle is $\frac{1}{2}|x - y|$ and the distance from its center to 0 is $\frac{1}{2}|x + y|$. Since the angle subtended at 0 by the diameter is less than $\pi/2$, the principal value of $\arg(x/y)$ is less than $\pi/2$ in magnitude.

Proof The definition (3) again lead to (4), since the principal value of $\log(x/y)$ is the value which tends to 0 as $x \rightarrow y$, that is, as $z \rightarrow 0$. From (4) we deduce

$$|A| < \frac{|z|^2}{3} (1 + |z|^2 + |z|^4 + \cdots) = \frac{|z|^2}{3(1 - |z|^2)},$$

which is equivalent to (8). $\square$

2.7 Stirling's formula for $n!$

If n is a large positive integer, say 100, then $n!$ is very large indeed and tedious to compute. An easy way of computing an approximate value was found by Stirling (1730) and modified by De Moivre, who showed that

$$n! \sim (2\pi n)^{1/2} (n/e)^n, \quad (1)$$

where e is the base of the natural logarithms. (The symbol $\sim$ means that the ratio of the two sides tends to unity as $n \rightarrow \infty$.) Even if n is as small as 10, the right side is only 1% less than the left. We shall prove Stirling's formula by showing first the connection between e and the Appell symbol $(\frac{1}{2}, n)$ for large values of n ,

$$(\frac{1}{2}, n) \sim 2^{1/2} (n/e)^n. \quad (2)$$

By multiplying this result and Wallis' formula (2.5-3) written in the form

$$n! / (\frac{1}{2}, n) \sim (\pi n)^{1/2}. \quad (3)$$

we obtain (1).

THEOREM 2.7-1 Let $n - 1 \in \mathbb{N}$. For some θ , depending on n , such that $0 < \theta < 1$,

$$\left(\frac{1}{2}, n\right) = 2^{1/2} n^n \exp\left(-n - \frac{\theta}{24n}\right). \quad (4)$$

Proof By the duplication theorem (2.2-7) we have, for any positive integer n ,

$$\begin{aligned} 2^{2n} \left(\frac{1}{2}, n \right) &= \frac{(1, 2n)}{(1, n)} = (n+1)(n+2) \cdots (2n-1)(2n) \\ &= \frac{1}{n^{n+1/2}} \left(\frac{n}{n+1} \right)^{n+1/2} \left(\frac{n+1}{n+2} \right)^{n+3/2} \cdots \left(\frac{2n-1}{2n} \right)^{2n-1/2} (2n)^{2n+1/2} \\ &= 2^{2n+1/2} n^n \prod_{k=n}^{2n-1} \left(\frac{k}{k+1} \right)^{k+1/2}. \end{aligned}$$

Thus, using (2.6-6),

$$\frac{2^{1/2} n^n}{\left(\frac{1}{2}, n \right)} = \prod_{k=n}^{2n-1} \left(\frac{k+1}{k} \right)^{k+1/2} = \exp \left(n + \frac{s}{12} \right), \quad (5)$$

$$0 < s < \sum_{k=n}^{2n-1} \frac{1}{k^2 + k}. \quad (6)$$

The last sum is a telescoping sum:

$$\sum_{k=n}^{2n-1} \frac{1}{k^2 + k} = \sum_{k=n}^{2n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n}.$$

Hence $s = \theta/2n$ for some θ such that $0 < \theta < 1$, which implies (4). $\square$

THEOREM 2.7-2 Let $n-1 \in \mathbb{N}$. Then

$$\lim_{n \rightarrow \infty} n! n^{-n-1/2} e^n = (2\pi)^{1/2}. \quad (7)$$

Proof By (4),

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}, n \right) n^{-n} e^n = 2^{1/2}. \quad (8)$$

Divide this equation by (2.5-3). $\square$

Equations (7) and (8) are equivalent to (1) and (2), respectively.

NOTES

A list of identities for Appell symbols is given by Slater (1966, Appendix I). For the theory of ${}_pF_q$ series see Erdélyi (1953, Chapter IV), Slater (1966), Bailey (1935), Luke (1969, 1975), and Rainville (1960, Chapter 5). Slater (1966, Sections 1.1, 2.1, and 3.1) gives some historical notes on hypergeometric series. Arithmetic and geometric means (of several values, with weights) are discussed by Beckenbach (1961), and the logarithmic mean by Carlson (1972a). A stronger version of Wallis' formula is given by Uppuluri (1966).

FORMULARY

$$(a, 0) = 1, \quad (a, n) = a(a+1) \cdots (a+n-1), \quad n-1 \in \mathbb{N}, \quad (2.2-2)$$

$$(1-z)^{-a} = \sum_{m=0}^{\infty} (a, m) \frac{z^m}{m!} = {}_1F_0(a; z), \quad |z| < 1, \quad (2.2-4)$$

$$(a, m+n) = (a, m)(a+m, n), \quad (2.2-5)$$

$$(a, -n) = \frac{(-1)^n}{(1-a, n)}, \quad (2.2-6)$$

$$(2a, 2n) = 2^{2n}(a, n)(a + \frac{1}{2}, n), \quad (2.2-7)$$

$$|(a, n)| \leq (|a|, n), \quad n \in \mathbb{N}, \quad (2.2-10)$$

$$\binom{a}{n} = (-1)^n \frac{(-a, n)}{n!}, \quad n \in \mathbb{N}, \quad (2.2-12)$$

$$\sum_{m=0}^n \frac{(a, m)(b, n-m)}{m!(n-m)!} = \frac{(a+b, n)}{n!}, \quad (2.3-1)$$

$$\sum_{\sum m_i = n} \frac{(b_1, m_1) \cdots (b_k, m_k)}{m_1! \cdots m_k!} = \frac{(b_1 + \cdots + b_k, n)}{n!}, \quad (2.3-5)$$

$$(x+y)^n = \sum_{m=0}^n \binom{n}{m} x^m y^{n-m}, \quad (2.3-8)$$

$$(x_1 + \cdots + x_k)^n = \sum_{\sum m_i = n} \frac{n!}{m_1! \cdots m_k!} x_1^{m_1} \cdots x_k^{m_k}, \quad (2.3-9)$$

$$n! \sim (2\pi n)^{1/2} (n/e)^n \sim (\pi n)^{1/2} \left(\frac{1}{2}, n\right), \quad n \rightarrow \infty. \quad (2.7-1,3)$$

EXERCISES

The number 2.2-1 designates the first of a series of exercises relating primarily to Section 2.2. Answers or hints for solution are given at the end of the book. Notations for sets are listed in Appendix A.

2.2-1 Let $n \in \mathbb{N}$, $r \in \mathbb{R}_+$, and $a \in \mathbb{C}$. If $|a| \geq r$, show that

$$|(a, n)/a^n| \leq (r, n)/r^n.$$

This inequality will be used in Section 5.10.

2.2-2 Let $n \in \mathbb{N}$ and $a \in \mathbb{C} - \{0, -1\}$. Express as ratios of Appell symbols the quantities $n+1$, $2n+1$, $4n+1$, $2n-1$, $4n^2-1$, $(n+a)/a$, $[n^2 + (2a+1)n + a(a+1)]/a(a+1)$.

2.2-3 If $n+1 \in \mathbb{N}$, the semifactorial function is defined by

$$\begin{aligned} n!! &= 1 \cdot 3 \cdot 5 \cdots (n-2)n, & n &= 1, 3, 5, \dots, \\ n!! &= 2 \cdot 4 \cdot 6 \cdots (n-2)n, & n &= 2, 4, 6, \dots, \\ 0!! &= (-1)!! = 1. \end{aligned}$$

If $m \in \mathbb{N}$ express $(2m)!!$, $(2m+1)!!$, and $(2m-1)!!$ in terms of Appell's symbol.

2.2-4 Let $m, n \in \mathbb{N}$ and $a \in \mathbb{C}$. Show that

$$(a, m-n) = \frac{(-1)^n(a, m)}{(1-a-m, n)} = \frac{(-1)^n(a-n, m)}{(1-a, n)}$$

whenever all members are well defined. Calculate each member separately for $a = \frac{1}{2}$, $m = 5$, $n = 3$.

2.2-5 Prove that $(1+n, n) = 4^n(\frac{1}{2}, n)$ if $n \in \mathbb{N}$. Why does the proof fail if $n < 0$? Find a similar expression for $(1+n, n)$ if $-n-1 \in \mathbb{N}$.

2.2-6 Let $k, n \in \mathbb{Z}$ and $a \in \mathbb{C}$. Provided that all Appell symbols are well defined, show that

- (i) $(-n, n) = (-1)^n n!$
- (ii) $(a, n) = (a-1, n) + n(a, n-1) = (a+1, n) - n(a+1, n-1)$,
- (iii) $(a-n, n) = (-1)^n(1-a, n)$,
- (iv) $(a+k, n-k) = (-1)^{n-k}(1-a-n, n-k)$,
- (v) $(a+n, k-n)(a+k, n-k) = 1$,
- (vi) $(a-2n, n) = (-1)^n(1-a+n, n)$,
- (vii) $(\frac{1}{2}+a-n, 2n) = (\frac{1}{2}-a-n, 2n)$.

2.2-7 If $f(m, n)$ is any function of m and n , show that

$$\sum_{m=0}^n \frac{f(m, n)}{(a, m)(b, n-m)} = \frac{(-1)^n}{(a, n)(b, n)} \sum_{m=0}^n (1-a-n, n-m)(1-b-n, m)f(m, n).$$

2.2-8 Let $n \in \mathbb{N}$ and $a \in \mathbb{C}$. Prove the triplication theorem,

$$(a, 3n) = 3^{3n} \left(\frac{a}{3}, n \right) \left(\frac{a+1}{3}, n \right) \left(\frac{a+2}{3}, n \right),$$

and more generally the multiplication theorem,

$$(a, kn) = k^{kn} \left(\frac{a}{k}, n \right) \left(\frac{a+1}{k}, n \right) \cdots \left(\frac{a+k-1}{k}, n \right), \quad k = 2, 3, 4, \dots$$

2.2-9 Prove that $|(a, n)| \geq |(\operatorname{Re} a, n)|$, where $n \in \mathbb{N}$ and $a \in \mathbb{C}$. If $\operatorname{Re} a \geq 0$, the second member can be replaced by $(\operatorname{Re} a, n)$.

2.3-1 Let $n \in \mathbb{N}$ and $b, c \in \mathbb{C}$, and assume $(c, n) \neq 0$. Prove that

$$\sum_{m=0}^n \frac{(-n, m)(b, m)}{(c, m)m!} = \frac{(c-b, n)}{(c, n)}.$$

This equation, often called Vandermonde's theorem, sums any terminating hypergeometric series (2.1-2) with unit argument.

2.3-2 Let $n \in \mathbb{N}$ and $p, q \in \mathbb{C}$. Prove that

$$\sum_{m=0}^n \binom{p}{m} \binom{q}{n-m} = \binom{p+q}{n}.$$

This identity is sometimes called Vandermonde's convolution.

2.3-3 If $a, b \in \mathbb{C}$, $n \in \mathbb{N}$, and $(a, n)(b, n) \neq 0$, show that

$$\sum_{m=0}^n \frac{1}{(a, m)(b, n-m)m!(n-m)!} = \frac{(a+b-1, n)}{(a, n)(b, n)n!}.$$

2.3-4 Sum the truncated binomial series with unit argument, $\sum_{m=0}^n (a, m)/m!$.

2.3-5 Show that

$$\sum_{m=0}^n \binom{n}{m}^2 = \binom{2n}{n}, \quad \sum_{m=0}^n \binom{2n}{2m} \binom{2m}{m} \binom{2n-2m}{n-m} = \binom{2n}{n}^2.$$

2.3-6 Evaluate $\sum (-1, m_1) \cdots (-1, m_k)$, where the sum extends over all nonnegative integers $m_1, \dots, m_k$ whose sum is n .

2.3-7 Let $n, s \in \mathbb{N}$ and $a + 1 \in U$. Express

$$\sum_{m=0}^n (-1)^{n-m} \binom{n}{m} \binom{a+m}{s}$$

as a single binomial coefficient.

2.3-8 If $p, q \in \mathbb{N}$, find the sum

$$\sum_{m=0}^p \sum_{n=0}^p \frac{(-1)^{m+n} (q+m+n)!}{(q+m)! (q+n)! (p-m)! (p-n)! m! n!}.$$

The problem arises in the quantum theory of an anisotropic harmonic oscillator [H. J. Rosenberg, *Amer. J. Phys.* **37** (1969), 839–840].

2.3-9 Let $n \in \mathbb{N}$ and $a \in \mathbb{C}$. Find the sum

$$\sum_{m=0}^n (-1)^m \binom{a+m-1}{m} \binom{a}{n-m}.$$

2.3-10 Let $n-1 \in \mathbb{N}$ and $a \in \mathbb{C}$. Show that

$$\sum_{m=0}^n (-1)^m \binom{a+m-1}{n-1} \binom{n}{m} = 0.$$

2.3-11 Let $n-1 \in \mathbb{N}$ and $a, b \in \mathbb{C}$. Find the sum

$$\sum_{m=0}^n m \frac{(a, m)(b, n-m)}{m! (n-m)!}.$$

2.3-12 Let n, p, q be positive integers. Prove Chu's theorem,

$$\sum_{r=1}^n \frac{(r, p)(n+1-r, q)}{p! q!} = \frac{(n, p+q+1)}{(1, p+q+1)}.$$

2.3-13 Let $M, N \in \mathbb{N}$ and $a, b \in \mathbb{C}$. Find the sum

$$\sum_{m=0}^M \sum_{n=0}^N \frac{(-1)^{m+n} (a+n, M)(b+m, N)}{m! (M-m)! n! (N-n)!}.$$

2.4-1 Let $z \in \mathbb{C}$ and $a, b \in U$, and assume $a+b \neq 1$. Show that

$${}_0F_1(a; z) {}_0F_1(b; z) = {}_2F_3\left(\frac{a+b-1}{2}, \frac{a+b}{2}; a+b-1, a, b; 4z\right)$$

and, in particular,

$$[J_0(z)]^2 = {}_1F_2\left(\frac{1}{2}; 1, 1; -z^2\right).$$

2.4-2 Let $n \in \mathbb{N}$ and $b, c \in \mathbb{C}$. Assume $(c, n) \neq 0$. Show that

$${}_2F_1(-n, b; c; 1) = \frac{(c-b, n)}{(c, n)}.$$

2.4-3 Let $n \in \mathbb{N}$ and $1 - \alpha - n \in U$. Show that

$${}_2F_0(-n, \alpha; z) = (\alpha, n)(-z)^n {}_1F_1(-n; 1 - \alpha - n; -1/z).$$

Passage from one side to the other is called reversal of the polynomial since the terms are taken in the reversed order. Show that the Hermite polynomial (2.4-19) can be written as a ${}_1F_1$ polynomial,

$$H_{2m}(z) = (-1)^m \frac{(2m)!}{m!} {}_1F_1(-m; \frac{1}{2}; z^2),$$

$$H_{2m+1}(z) = (-1)^m \frac{(2m+1)!}{m!} 2z {}_1F_1(-m; \frac{3}{2}; z^2).$$

Show also that the Laguerre polynomial (2.4-18) can be written as a ${}_2F_0$ polynomial,

$$L_n^\alpha(z) = \frac{(-z)^n}{n!} {}_2F_0(-n, -n - \alpha; -1/z).$$

2.4-4 Let $p, q, n \in \mathbb{N}$, $(a) = (a_1, \dots, a_p)$, $(b) = (b_1, \dots, b_q)$, $(1 - a - n) = (1 - a_1 - n, \dots, 1 - a_p - n)$, $(1 - b - n) = (1 - b_1 - n, \dots, 1 - b_q - n)$. Assume $(b) \in U^q$ and $(1 - a - n) \in U^p$. Prove the reversal formula,

$${}_{p+1}F_q[-n, (a); (b); z] = (-z)^n \frac{(a_1, n) \cdots (a_p, n)}{(b_1, n) \cdots (b_q, n)} {}_{q+1}F_p[-n, (1 - b - n); (1 - a - n); (-1)^{p+q} z^{-1}].$$

2.4-5 Let $m, n \in \mathbb{N}$ and $c_1, \dots, c_q \in U$. Define

$$u_m = \frac{(a_1, m) \cdots (a_p, m) z^m}{(c_1, m) \cdots (c_q, m) m!}.$$

Show that the truncated ${}_pF_q$ series is expressible in terms of ${}_{p+1}F_{q+1}$ by

$$\sum_{m=n}^{\infty} u_m = u_n {}_{p+1}F_{q+1}(1, a_1 + n, \dots, a_p + n; 1 + n, c_1 + n, \dots, c_q + n; z).$$

2.4-6 Define the differential operator $\delta = z(d/dz)$. Show that $y = {}_pF_q[(a), (c), z]$ satisfies (inside its circle of convergence) the linear differential equation

$$z \prod_{i=1}^p (\delta + a_i) y = \delta \prod_{j=1}^q (\delta + c_j - 1) y.$$

2.4-7 Let $n \in \mathbb{N}$, $(a) = (a_1, \dots, a_p)$, $(c) = (c_1, \dots, c_q)$, $(a + n) = (a_1 + n, \dots, a_p + n)$, and $(c + n) = (c_1 + n, \dots, c_q + n)$. Assume $(a) \in \mathbb{C}^p$ and $(c) \in U^q$. Show that the n th derivative of a ${}_pF_q$ series (inside its circle of convergence) is

$$\frac{d^n}{dx^n} {}_pF_q[(a), (c), x] = \frac{(a_1, n) \cdots (a_p, n)}{(c_1, n) \cdots (c_q, n)} {}_pF_q[(a + n), (c + n), x].$$

The case $p = 0, q = 1$ will figure in (6.9-23).

2.4-8 The Kelvin functions, $\text{ber}(x)$ and $\text{bei}(x)$ (in which “be” comes from Bessel and the third letter stands for “real” or “imaginary”), occur as real and imaginary parts of J_0 on a line which makes an angle of 45 degrees with the real axis. They are encountered, for example, in the theory of the skin effect in transmission of high-frequency current through

a wire of circular cross section. They are defined by

$$\text{ber}(x) \pm i \text{bei}(x) = J_0(xe^{\pm i3\pi/4}).$$

Show that

$$\text{ber}(x) = {}_0F_3\left(1, \frac{1}{2}, \frac{1}{2}; -x^4/256\right),$$

$$\text{bei}(x) = (x^2/4) {}_0F_3\left(1, \frac{3}{2}, \frac{3}{2}; -x^4/256\right),$$

and, more generally,

$${}_0F_1(c; z) = {}_0F_3\left(\frac{1}{2}, \frac{c}{2}, \frac{c+1}{2}; \frac{z^2}{16}\right) + \frac{z}{c} {}_0F_3\left(\frac{3}{2}, \frac{c+1}{2}, \frac{c+2}{2}; \frac{z^2}{16}\right).$$

2.4-9 Let $n \in \mathbb{N}$, $(a) = (a_1, \dots, a_p)$, $(c) = (c_1, \dots, c_q)$, $(1-a-n) = (1-a_1-n, \dots, 1-a_p-n)$, and $(1-c-n) = (1-c_1-n, \dots, 1-c_q-n)$, with similar definitions for primed quantities. Assume $(c), (c') \in U^q$ and $(1-a-n) \in U^p$. Prove the following equality between formal power series (to be used in Section 7.2):

$$\begin{aligned} & {}_pF_q[(a), (c), x] {}_pF_q[(a'), (c'), y] \\ &= \sum_{n=0}^{\infty} \frac{(a_1, n) \cdots (a_p, n) x^n}{(c_1, n) \cdots (c_q, n) n!} {}_{p+q+1}F_{p+q} \left[\begin{matrix} -n, (a'), (1-c-n) \\ (c'), (1-a-n) \end{matrix}; (-1)^{p+q+1} (y/x) \right]. \end{aligned}$$

2.4-10 Let $a, b, b', c, x, y \in \mathbb{C}$ and assume $|x| < 1$, $|y| < 1$, and $c \in U$. Prove that the double series (2.4-25) converges absolutely.

2.5-1 Prove that

$$\prod_{n=1}^{\infty} (1 - 1/4n^2) = 2/\pi.$$

2.5-2 Let $n-1 \in \mathbb{N}$. Prove that

$$\lim_{n \rightarrow \infty} \frac{(n!)^2 2^{2n}}{(2n)! n^{1/2}} = \pi^{1/2}.$$

2.5-3 Prove that

$$\prod_{n=1}^{\infty} \left[1 + \frac{1}{(4n-1)(4n-3)} \right] = 2^{1/2}.$$

2.5-4 Let $n-1 \in \mathbb{N}$. For some θ , depending on n , such that $0 < \theta < 1$, show that

$$\frac{n!}{(\frac{1}{2}, n)} = (\pi n)^{1/2} \left(1 + \frac{\theta}{4n} \right).$$

2.6-1 Let $x, u \in \mathbb{R}_{>}^k$ and assume $\sum_{i=1}^k u_i = 1$. Prove the inequality of the weighted arithmetic and geometric means,

$$\prod_{i=1}^k x_i^{u_i} \leq \sum_{i=1}^k u_i x_i,$$

with strict inequality unless all the x_i are equal.

2.6-2 In (2.6-6) show that $0 < \theta < \frac{1}{6}$.

2.6-3 If $x, y \in \mathbb{C}$ show that the geometric and logarithmic means of x and y are not always less in absolute value than the arithmetic mean.

2.6-4 Let $x, y \in \mathbb{C}$. Show that

$$\left| \frac{x+y}{2} + (xy)^{1/2} \right| + \left| \frac{x+y}{2} - (xy)^{1/2} \right| = |x| + |y|.$$

Show further that the terms on the left are $a \pm b$, where a and b are lengths of semiaxes of the ellipse with foci x and y which passes through 0.

2.6-5 Let $\theta \in \mathbb{C}$. Show that

$$|\cos \theta + 1| + |\cos \theta - 1| = 2 \cosh \operatorname{Im} \theta.$$

2.6-6 Let $x, y \in \mathbb{C}$. Assume that $x \neq y$ and the line through x and y does not pass through 0. Show that 0 and one value of $(xy)^{1/2}$ lie on opposite sides of that line.

2.7-1 Let $n-1 \in \mathbb{N}$. For some θ , depending on n , such that $0 < \theta < 1$, show that

$$\left(\frac{1}{2}, n \right) = 2^{1/2} n^n \exp \left(-n - \frac{1}{24n} + \frac{\theta}{192n^3} \right).$$

2.7-2 Let $n-1 \in \mathbb{N}$. For some φ , depending on n , such that $-1 < \varphi < 5$, show that

$$n! = (2\pi)^{1/2} n^{n+1/2} \exp(-n + \varphi/24n).$$